Global Business Intelligence & Customer Analytics Platform (Olist E-commerce Dataset)

1. Project Overview

- This project builds an end-to-end business intelligence system using the Olist e-commerce dataset.
- Focus areas include sales performance, customer behavior, logistics efficiency, and satisfaction.
- The project simulates a real-world analytics environment with multiple datasets.

2. Problem Statement

- E-commerce data is distributed across multiple tables (orders, customers, payments, etc.).
- Businesses struggle to derive insights without integrating these datasets.
- This project solves this by building a unified analytics platform.

3. Objectives

Business Objectives

- Understand revenue and sales trends
- Analyze customer purchasing behavior
- Evaluate seller and product performance
- Improve delivery and customer satisfaction

Technical Objectives

- Perform data cleaning and transformation using Python
- Design relational database schema using SQL
- Create dashboards using Power BI
- Generate business insights and KPIs

4. Dataset Description

- Orders: order lifecycle and timestamps
- Order Items: product-level details
- Payments: payment methods and values

- Customers: demographics and location
- Products: product categories
- Reviews: customer satisfaction scores
- Sellers: seller information
- Geolocation: location data

5. Tech Stack

- Python (Pandas, NumPy) for data cleaning and analysis
- SQL (PostgreSQL/MySQL) for database management
- Power BI for visualization and dashboards
- Excel for quick analysis and validation
- GitHub for version control

6. Data Workflow

- Load raw CSV data into Python
- Clean and preprocess data
- Store processed data into SQL database
- Perform analysis using SQL and Python
- Visualize insights in Power BI
- Document findings in report

7. Data Modeling

- Fact Table: Order Items (central table)
- Dimension Tables: Customers, Products, Sellers, Payments, Date
- Use primary and foreign keys to connect tables
- Build star schema for efficient querying

8. Key KPIs

- Total Revenue
- Total Orders
- Average Order Value
- Customer Retention Rate
- Delivery Time
- Delayed Orders Percentage
- Average Review Score

9. Power BI Dashboard

- Executive Overview Dashboard
- Customer Insights Dashboard
- Product and Seller Performance Dashboard
- Operations and Delivery Dashboard
- Customer Satisfaction Dashboard

10. Advanced Analytics (Optional)

- Customer Segmentation using clustering
- RFM analysis
- Cohort analysis
- Review prediction models

11. Deliverables

- Python notebooks
- SQL schema and queries
- Power BI dashboard
- Excel analysis file
- Project report
- GitHub repository

12. Conclusion

- This project demonstrates end-to-end data analytics capabilities.
- It showcases real-world problem-solving and business insight generation.
- Highly valuable for portfolio and job applications.